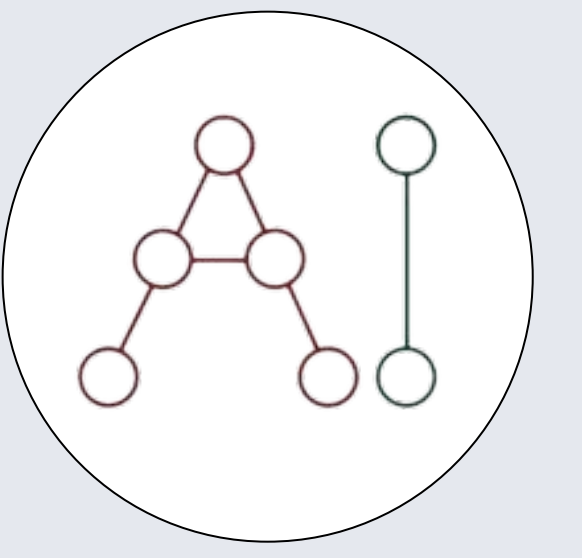




MASON

Gerry Agluba Jr
Miguel Remolona, PhD (Adviser)

A MULTI-AGENT HARNESS FOR SERVICE ORCHESTRATION AND OPTIMIZATION



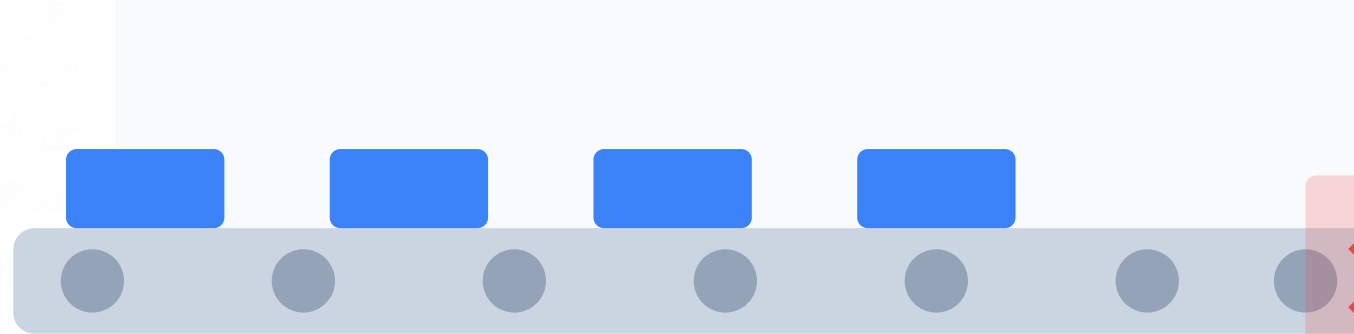
INTRODUCTION

PROBLEM

- Standard pipelines for service lifecycle management **fail to adapt to unpredictable real-world events**, resulting in **slow incident triage** and **inconsistent remediation**.
- Tool Fragmentation has caused **cognitive burden** for Engineers



AUTOMATION FAILURE



Constrained / Dynamic Environment

- Network Policies
- Resource Quotas
- Infra / Platform Changes

We need an **intelligent** orchestration agent that is **non-disruptive**, **conversational**, **extensible**, **autonomous** and can handle unforeseen scenarios with the **capability to improve**

USER

Alex
Newbie Engineer

Alex: Resolve issues with failed services and create an incident diagnosis and resolution report.

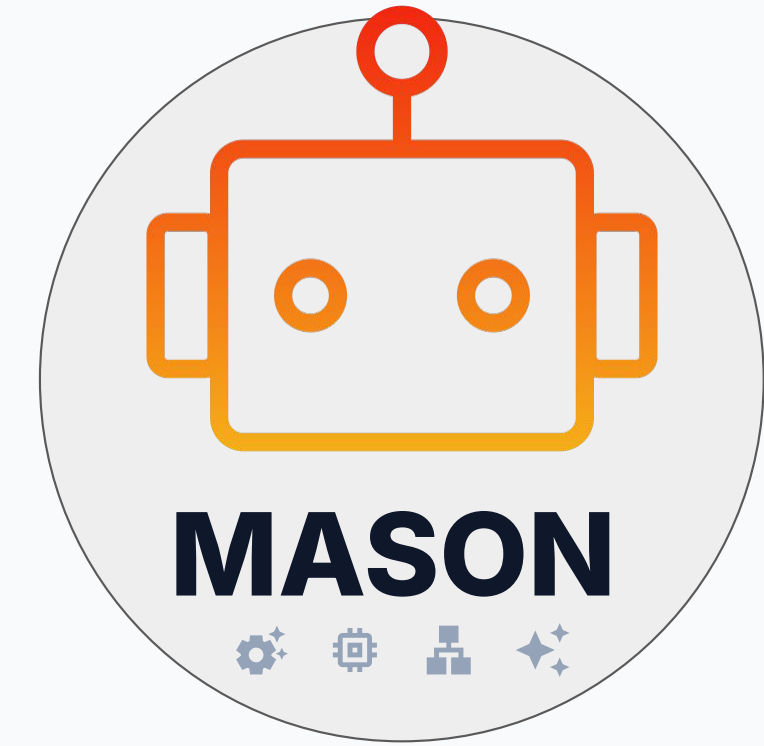
USER

Mike
Senior Engineer

Alex: Perform an agent optimization on the deployer.

EVENTS

A New Package is Release, Perform Deployment



ORCHESTRATION
(Cloud-native context-thrifty tools)

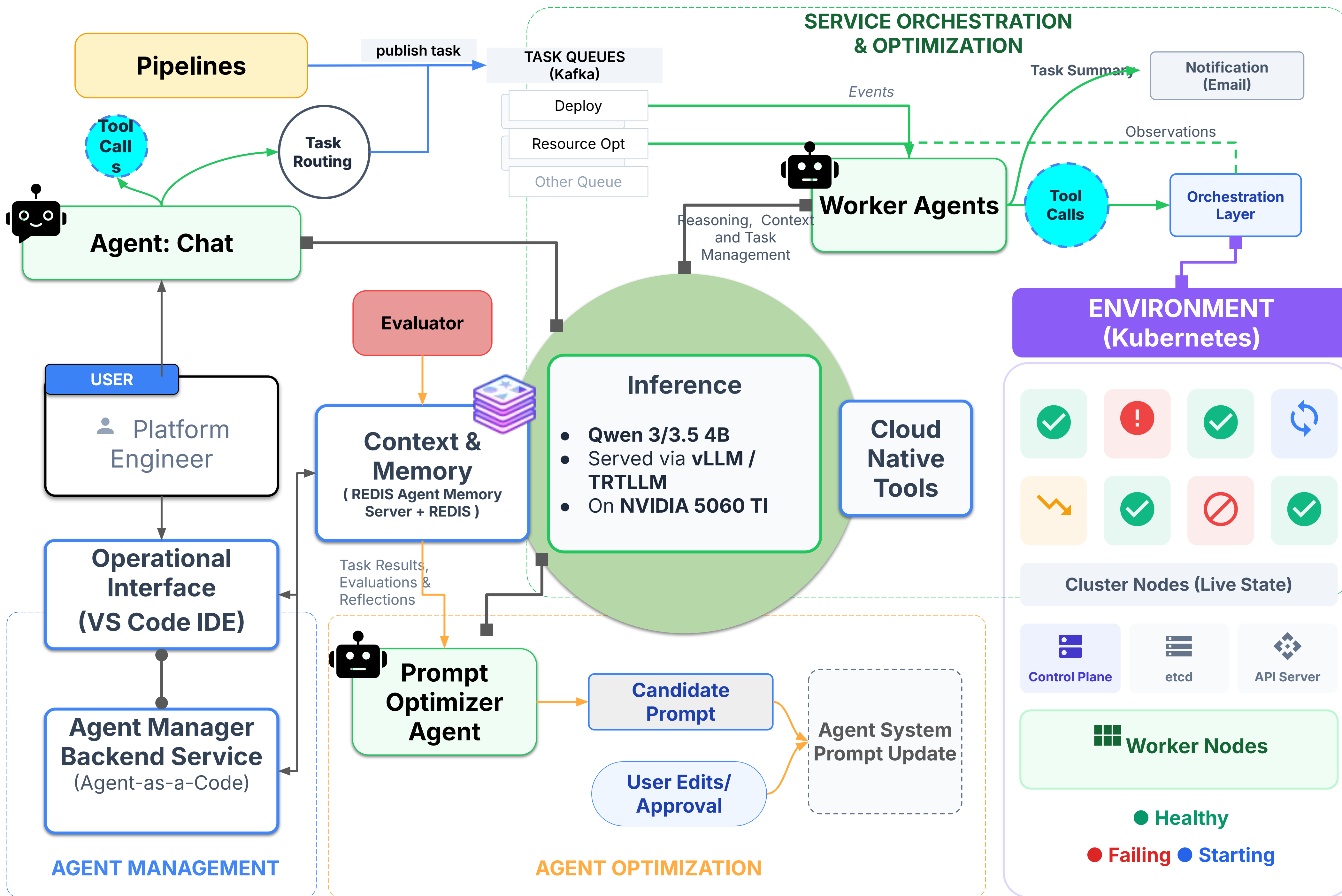
AGENT OPTIMIZATION
(User-Assisted Prompt Optimization)

FRAMEWORK
(Agents-as-Code)

LLM ENGINE
($\leq 5B$ Params)

METHODOLOGY

SYSTEM ARCHITECTURE AND OPERATION



UNIFIED OPERATIONAL INTERFACE

To resolve tool fragmentation and complex operations. An **IDE-based interface** is developed for targeted users using the system for

- (a) Conversational Tasks,
- (b) Viewable Reports,
- (c) User-Assisted Prompt Optimization
- (D) Agent Lifecycle Management.



Scan QR Code For Video Demo of System Operation

EVALUATION

Scenario Creation

Agentic Use Cases

- Deployment: *Diff. preconditions, happy paths, edge cases.*
- Ad-Hoc: *Resource inventories, workload analysis, recovery, policy friction.*

Structure

- Pre-Condition • Execution • Post-Condition

LLM as a Judge

Judge (Batch API Calls)

- gemini-3.1-flash-lite-preview**
- GPT 5.4**

Inputs

Definitions • Prompt • Evidences • Available Tools

Outputs

Evaluation Scores & Remarks

RESULTS

Setup

- Setup A - Qwen 3 Instruct 4B served on TensorRT-LLM
- Setup B - Qwen 3.5 4B served on vLLM

Task Type	Completion on Gemini Eval (Setup A vs Setup B)	Completion on GPT Eval (Setup A vs Setup B)
Deployment	80% vs 92%	72% vs 81%
Ad-Hoc	78% vs 88%	72% vs 88%

Agent Metrics over Setup (GPT Eval)	Ad-Hoc		Deployment	
	Setup A	Setup B	Setup A	Setup B
Tool Accuracy	0.7	0.8	0.74	0.81
Step Efficiency	0.71	0.79	0.69	0.82
Plan Adherence	0.71	0.81	0.63	0.78
Faithfulness	0.67	0.79	0.62	0.71
Operational Efficiency				
Workflow Latency (s)	31.4986	93.6711	140.1829	217.4672
Function Calls	4.88	6.72	7.88	13.6
Generated Token Cost (tokens)	297.6	1540.16	994.8	2915.76
TTFT (s)	4.0248	2.6683	2.8923	1.6507

DISCUSSION

- High Performance**
 - Up **92%** deployment and **88%** *ad-hoc task task completions*
- Precision vs Operational Efficiency**
 - Qwen 3.5 4B ~80%** vs Qwen 3 Instruct 4B 's 60%-70% **across metrics** highlights Agents's inherent necessity on **Qwen'sthinking mode**; albeit the latter model's **reduction** on workflow *latency, token costs* and *tool calls* by **66%, 80.6%** and **42%** respectively.
- Fast Evolution**
 - The Prompt-Optimizer accelerated ad-hoc task completion rates from **52% to 76% in just 3 optimization steps.**

CONCLUSION

MASON's self-optimizing orchestration layer provides a precise and low-cost solution for managing complex cloud-native environments. Additionally, it eliminates tool fragmentation and minimizes human intervention for resilient, cost-effective, and autonomous cloud operations.